AI Agents for Resource Allocation in 6G Networks: A Harness Engineering Perspective

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***Abstract*—With the rapid development of large language models (LLMs), the integration of LLMs with specialized domains is becoming an important trend. In particular, integrating LLMs with wireless communications offers a promising solution for realizing AI-native networks in 6G. Existing studies mainly focus on improving the LLMs themselves, for example, through prompt engineering, retrieval-augmented generation (RAG), and fine-tuning to better adapt them to specific communication tasks. However, relying solely on model adaptation cannot fully overcome the inherent limitations of LLMs in engineering optimization, such as hallucination, numerical inaccuracy, and weak verifiability. To this end, inspired by harness engineering, we develop WirelessNetAgent for resource allocation in next-generation wireless networks. The core idea is to construct a domain-specific support system around a general-purpose LLM, rather than adapting the LLM to network optimization. Within the support system, we design an agent workflow, a callable optimization tool, and an execution environment based on digital twin, enabling a closed-loop process from natural-language intent parsing to resource allocation and performance verification. Simulation experiments verify that the proposed WirelessNetAgent can effectively parse complex intents expressed in natural language and optimize the network accordingly.**

***Index Terms*—AI agent, digital twin, harness engineering, large language model, network optimization, resource allocation.**

I. Introduction

The 6G wireless networks are envisioned to deliver transformative capabilities, including terabit-level throughput, sub-millisecond latency, and massive connectivity [1]. Advanced architectures such as cell-free (CF) massive multiple-input multiple-output (MIMO) [2] are expected to play an important role in enabling these capabilities. Different from conventional cellular systems with fixed cell boundaries, CF networks allow a large number of distributed access points (APs) to coherently serve user equipment (UEs) over the same time-frequency resources. By exploiting macro diversity and spatial degrees of freedom, they can enhance spectral efficiency (SE), improve service reliability, and enable scalable connectivity in future 6G networks. These advantages are essential for supporting emerging applications such as immersive extended reality (XR), the low-altitude economy (LAE), and the industrial Internet of things (IIoT). Nevertheless, these applications impose highly heterogeneous quality of service (QoS) requirements, substantially increasing the complexity of resource allocation in CF networks.

In this context, static rule-based paradigms can hardly cope with the diverse service requirements in 6G systems, while manual configuration is expertise-intensive and time-consuming. AI-based methods have been extensively investigated for wireless network resource allocation, and have demonstrated remarkable performance. However, most existing methods are developed for predefined problem formulations, which limits their flexibility in adapting to various problems. Specifically, these AI methods are usually optimized with loss functions designed for a given objective, such as maximizing system throughput or enhancing user fairness. Meanwhile, optimization constraints, for instance avoiding performance degradation or guaranteeing minimum rate thresholds, are often incorporated through dedicated mechanisms. As a result, when the optimization objective or constraint set changes, the learned model may no longer be directly applicable and often requires retraining or redesign.

A. From LLMs to LLM Agents in Wireless Communications

Since OpenAI® released ChatGPT in November 2022, large language models (LLMs) have rapidly attracted broad attention from both academia and industry, and have been progressively integrated with various research domains. In the field of wireless communications, WirelessLLM [3] presents a comprehensive framework tailored to wireless systems, which incorporates specific wireless knowledge into LLMs and demonstrates its effectiveness across representative tasks, including power allocation, spectrum sensing, and protocol understanding. LLM4CP [4] introduces pre-trained GPT-2 into the channel prediction task and fine-tunes the LLM to predict the future downlink channel state information (CSI) based on the historical uplink CSI. In [5], an LLM-based combinatorial optimizer (LMCO) is developed for wireless network planning, where LLMs are guided by prompts and expert propagation knowledge to determine AP numbers and placements. In [6], the authors propose an LLM-empowered resource allocation framework for intent-driven wireless networks, enabling adaptive and numerically precise resource allocation for wireless federated learning and reconfigurable intelligent surface (RIS)-assisted unmanned aerial vehicle (UAV) communications.

The source code is publicly available at https://github.com/yangbin-xd/cell-free.git. An interactive demo is available at https://huggingface.co/spaces/Bin-Yang/cell-free.

However, despite the strong capabilities of LLMs, directly applying them to practical wireless communication is still subject to several limitations, as follows.

- LLMs may suffer from hallucinations [7], and harness engineering [8] is needed to guide, constrain, and validate their behavior through structured workflows.
- LLMs are not inherently reliable for precise numerical optimization [9], and therefore require appropriate external tools according to the specific problem context.
- Standalone LLMs lack an inherent closed-loop feedback mechanism, making it necessary to construct an executable environment in which the generated strategies can be executed and further validated before deployment.

These limitations motivate the development of LLM agents [10], which extend standalone LLMs from passive language models to complete autonomous systems capable of hallucination mitigation, tool invocation, and solution verification. ComAgent [11] coordinates multiple LLM agents for literature search, planning, coding, and scoring, transforming wireless optimization intents into executable and verifiable simulation pipelines. WirelessAgent [12] builds an agentic framework with perception, memory, planning, and action modules, enabling interactions with external tools and execution of network slicing tasks. In [13], a split learning system is proposed for LLM agents in 6G networks, where agents collaborate to support multi-modal interaction and task execution.

B. Contributions and Paper Organizations

Motivated by the above discussions, this paper develops WirelessNetAgent, an LLM agent that enables resource allocation under diverse optimization objectives described in natural language. The key innovation of this work lies in incorporating harness engineering into the developed framework for wireless network optimization. The main contributions are listed below.

- We design an agent workflow that integrates context engineering, field validation, tool invocation, and performance evaluation in an execution environment to ensure reliable task completion by the developed agent.
- We develop an agent callable optimization tool for CF network resource allocation, supporting both power allocation with fixed topology and joint AP-UE association and power allocation.
- We construct an execution environment for CF networks, enabling the agent to verify candidate solutions with reliable performance feedback before deployment.

The remainder of this paper is organized as follows. Section II presents the system model of the considered CF network. Section III introduces the framework of the proposed WirelessNetAgent, including the designed agent workflow, agent callable optimization tools, and the constructed digital twin for CF networks. Section IV provides the numerical simulation results, and Section V concludes this paper.

II. System Model

We consider a CF massive MIMO system, where a central controller (CC) coordinates a total of L APs to cooperatively serve K UEs over the same time-frequency resources. The system operates in time-division duplex (TDD) mode, enabling the APs to acquire downlink channel state information (CSI) from uplink pilots through channel reciprocity. Each AP is equipped with a uniform planar array (UPA) comprising $N = N_x \times N_y$ antennas and applies maximum-ratio transmission (MRT) beamforming based on the estimated CSI, while each UE is equipped with a single antenna. Orthogonal frequency division multiplexing (OFDM) is adopted, with the system bandwidth and the number of subcarriers denoted by B and M, respectively. For scalability, each UE selects only a subset of nearby APs rather than being served by all APs.

Let $\mathcal{A} = \{a_l \mid l = 1, \cdots, L\}$ and $\mathcal{U} = \{u_k \mid k = 1, \cdots, K\}$ denote the sets of APs and UEs, respectively. The AP-UE association is represented by a binary matrix $\mathbf{B} = [b_{l,k}] \in \{0,1\}^{L \times K}$, where $b_{l,k} = 1$ indicates that a_l serves u_k, and 0 otherwise. The downlink power allocation is described by $\mathbf{P} = [p_{l,k}] \in \mathbb{R}^{L \times K}$, where $p_{l,k}$ denotes the transmit power assigned from a_l to u_k. The set of APs serving u_k is denoted by $\mathcal{A}(k) = \{l \mid b_{l,k} = 1\}$, and the set of UEs served by a_l is denoted by $\mathcal{U}(l) = \{k \mid b_{l,k} = 1\}$. With this notation, the transmitted symbol of a_l on the m-th subcarrier is given by

$$\mathbf{x}_l^m = \sum_{k \in \mathcal{U}(l)} \sqrt{p_{l,k}^m} \mathbf{w}_{l,k}^m s_k^m, \tag{1}$$

where $\mathbf{w}_{l,k}^m \in \mathbb{C}^{N \times 1}$ is the precoding vector applied at a_l for u_k over the m-th subcarrier, and s_k^m is the corresponding data symbol. Then, the frequency-domain observation of the received signal at u_k over the m-th subcarrier is expressed as

$$\begin{aligned} y_k^m &= \sum_{l=1}^{L} \mathbf{h}_{l,k}^{m\,H} \mathbf{x}_l^m + n_k^m \\ &= \underbrace{\sum_{l \in \mathcal{A}(k)} \sqrt{p_{l,k}^m}\, \mathbf{h}_{l,k}^{m\,H} \mathbf{w}_{l,k}^m s_k^m}_{\text{Signal}} + \\ &\quad \underbrace{\sum_{k' \neq k} \sum_{l \in \mathcal{A}(k')} \sqrt{p_{l,k'}^m}\, \mathbf{h}_{l,k}^{m\,H} \mathbf{w}_{l,k'}^m s_{k'}^m}_{\text{Interference}} + n_k^m, \end{aligned} \tag{2}$$

where $\mathbf{h}_{l,k}^m \in \mathbb{C}^{N \times 1}$ denotes the channel from a_l to u_k over the m-th subcarrier, $n_k^m \sim \mathcal{CN}(0, \sigma^2)$ is the complex additive Gaussian noise.

Different services carried by different UEs impose heterogeneous QoS requirements on the wireless network, including data rate, reliability, and latency. In this work, we focus on the rate requirements from different UEs and apply SE as the performance metric. Specifically, the SE of u_k is averaged over all subcarriers as

$$R_k = \frac{1}{M} \sum_{m=1}^{M} \log_2\left(1 + \frac{S_k^m}{I_k^m + \sigma^2}\right), \tag{3}$$

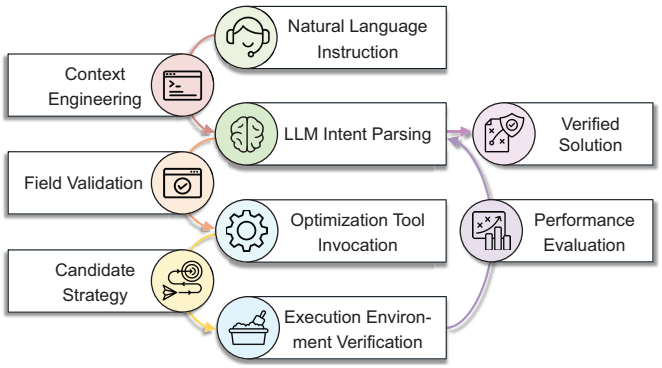


Fig. 1. The workflow of WirelessNetAgent.

where S_k^m represents the received signal power of u_k on the m-th subcarrier, given by

$$S_k^m = \left| \sum_{l \in \mathcal{A}(k)} \sqrt{p_{l,k}^m}\, \mathbf{h}_{l,k}^{m\,H} \mathbf{w}_{l,k}^m \right|^2, \tag{4}$$

and I_k^m denotes the corresponding interference power, calculated as

$$I_k^m = \sum_{k' \neq k} \left| \sum_{l \in \mathcal{A}(k')} \sqrt{p_{l,k'}^m}\, \mathbf{h}_{l,k}^{m\,H} \mathbf{w}_{l,k'}^m \right|^2. \tag{5}$$

III. The Framework of WirelessNetAgent

The key idea of WirelessNetAgent is to introduce a harness around the LLM. The harness refers to the supporting infrastructure that surrounds the model and enables it to operate as an agent for practical task execution [8]. An intuitive analogy is that, if the LLM is regarded as a *horse*, then the harness can be viewed as the *tack* that guides, constrains, and controls the horse. Their integration transforms the standalone model into an agent capable of reliable task execution. In practice, even a powerful model may fail without a stable and controllable harness. Therefore, harness engineering plays a critical role in bridging the gap between the reasoning capability of the LLM and reliable real-world operation.

To realize this bridge in wireless network optimization, this section presents the framework design of WirelessNetAgent, which operationalizes harness engineering through three key components: an agent workflow, callable optimization tools, and an execution environment.

A. The Agent Workflow

A standalone language model usually performs a single forward pass after receiving an input and then terminates, which limits its ability to solve complex problems in practice. Therefore, the agent workflow constitutes the operational core of the agentic harness, transforming a stateless language model into a task-oriented system. As for the resource allocation in CF networks, a dedicated workflow is required to transform natural language service requirements into structured optimization objectives and constraints, and to generate reliable resource allocation strategies.

The designed workflow is illustrated in Fig. 1. It starts with natural language instructions, which are a natural form for describing user service requirements in practical wireless networks. For example, some users may require high-definition video conferencing, while others may only require voice communication. These heterogeneous service demands correspond to different rate requirements. The agent first collects the current natural language instruction $\boldsymbol{q}_t$ and integrates it with the historical dialogue, given by

$$\boldsymbol{D}_{t-1} = \big\{(\boldsymbol{q}_1, \boldsymbol{g}_1), (\boldsymbol{q}_2, \boldsymbol{g}_2), \cdots, (\boldsymbol{q}_{t-1}, \boldsymbol{g}_{t-1})\big\}, \tag{6}$$

where $\boldsymbol{q}_i$ and $\boldsymbol{g}_i$ denote the user input and the LLM output at the i-th interaction round, respectively. The current network state $\boldsymbol{s}_t$ is also incorporated to provide real-time information of the network including the AP-UE association matrix $\mathbf{B}$, the allocated power matrix $\mathbf{P}$, and the achieved user rates $\mathbf{r} = [R_1, \cdots, R_K]^T \in \mathbb{R}^{K \times 1}$ of all K UEs, i.e.,

$$\boldsymbol{s}_t = \big\{\mathbf{B}, \mathbf{P}, \mathbf{r}\big\}. \tag{7}$$

In addition, the system prompt $\boldsymbol{p}$ designed for the resource allocation task is introduced to construct the current context input as

$$\boldsymbol{c}_t = \Phi\big(\boldsymbol{p}, \boldsymbol{s}_t, \boldsymbol{D}_{t-1}, \boldsymbol{q}_t\big), \tag{8}$$

where $\Phi(\cdot)$ represents the context engineering function that assembles these information blocks into a unified prompt. Based on $\boldsymbol{c}_t$, the LLM performs intent parsing and produces a structured task description as

$$\boldsymbol{g}_t = \Psi\big(\boldsymbol{c}_t\big), \tag{9}$$

where $\Psi(\cdot)$ denotes the intent parsing function. The output $\boldsymbol{g}_t$ contains the parsed optimization objective $\mathbf{r}^*$, constraint set $\mathcal{C}$, and task-specific parameter w, i.e.,

$$\boldsymbol{g}_t = \big\{\mathbf{r}^*, \mathcal{C}, w\big\}, \tag{10}$$

where $\boldsymbol{r}^* = [R_1^*, \cdots, R_K^*]^T \in \mathbb{R}^{K \times 1}$ denotes the target rates of all K UEs, and w is a scalar indicator that specifies whether the current network topology can be modified. The parsed output $\boldsymbol{g}_t$ is typically organized in JSON format as specified by the system prompt $\boldsymbol{p}$.

Nevertheless, $\boldsymbol{g}_t$ may still contain invalid values, unsupported objectives, or infeasible constraints. Therefore, field validation is performed before tool invocation to ensure that the structured task description satisfies the predefined schema. After that, the validated task description is passed to the tool invocation module, which employs the designed optimization tool to solve the task according to the parsed output. In this work, we design a unified agent callable optimization tool based on gradient descent. Depending on the parsed task-specific parameter w, the tool can perform power allocation while keeping the current topology unchanged, or jointly optimize the AP selection and power allocation.

Next, the candidate strategy generated by the selected tool is evaluated in the execution environment, which is constructed based on a digital twin of the physical CF network. This step verifies whether the candidate strategy is feasible and whether it satisfies the user requirements under the current network status. Specifically, the agent applies the candidate strategy to the execution environment and returns the predicted user rates $\hat{\mathbf{r}}$. The evaluation results $\hat{\mathbf{r}}$ are then fed back to the LLM. Based on the original user instruction and the parsed task description, the LLM determines whether the candidate strategy satisfies the required objective and constraints. If the evaluation results meet the user requirements, the agent returns the candidate strategy as the verified solution. Otherwise, the LLM uses the feedback to refine the task interpretation, or adjust the optimization parameters to initiate another loop.

B. Optimization Tools and Execution Environment

In the designed workflow, the agent is required to generate executable resource allocation strategies according to user specified requirements. This process depends not only on the optimization tool, but also on an execution environment that can accurately evaluate the candidate strategies. Therefore, the optimization tool and execution environment constitute key components of the workflow of WirelessNetAgent.

However, both components face inherent challenges. For the optimization tool, resource allocation in CF networks is highly complex. Different user requirements may correspond to different optimization objectives, such as maximizing system throughput, improving user fairness, or enhancing the data rates of specific users. Therefore, it is difficult to develop an optimization tool that can effectively handle diverse objectives and constraints across different CF resource allocation tasks. For the execution environment, the core task is to construct the mapping from the CSI $\mathbf{H} \in \mathbb{C}^{L\times K\times N_t\times M}$, association matrix $\mathbf{B}$, and power allocation matrix $\mathbf{P}$ to the achievable user rates $\mathbf{r}$, i.e.,

$$\mathbf{r} = \mathcal{F}(\mathbf{H}, \mathbf{B}, \mathbf{P}), \tag{11}$$

where $\mathcal{F}(\cdot)$ represents the achievable rates mapping defined by (3). Although instantaneous CSI $\mathbf{H}$ can be obtained in practical TDD systems through uplink pilot transmission and channel reciprocity, using it as the direct input of the execution environment is costly. In CF networks, the CSI dimension grows rapidly with the numbers of APs L, users K, AP antennas N_t, and subcarriers M. The frequent acquisition, update, and transmission of such high dimensional CSI introduce considerable pilot overhead, communication latency, and computational burden. Alternatively, CSI can be generated through ray tracing, but this also requires repeated electromagnetic propagation simulation for different user locations, which is computationally expensive and thus unsuitable for fast strategy evaluation.

Based on radio map technology [14], we construct a differentiable digital twin for CF networks to address both challenges simultaneously. Radio map is a promising technology that captures the relationship between UEs' locations and their historical CSI records, enabling the inference of channel information solely from UEs' positions without pilot measurements [15]. Therefore, by exploiting radio map, the high dimensional CSI input $\mathbf{H}$ in (11) can be replaced by low-dimensional UE locations $\mathbf{X}$, where $\mathbf{X} \in \mathbb{R}^{K\times 2}$ denotes the positions of all K UEs. The constructed digital twin is shown in Fig. 2, which establishes the mapping from UE locations $\mathbf{X}$, association matrix $\mathbf{B}$, and power allocation matrix $\mathbf{P}$ to the predicted user rates $\hat{\mathbf{r}}$, i.e.,

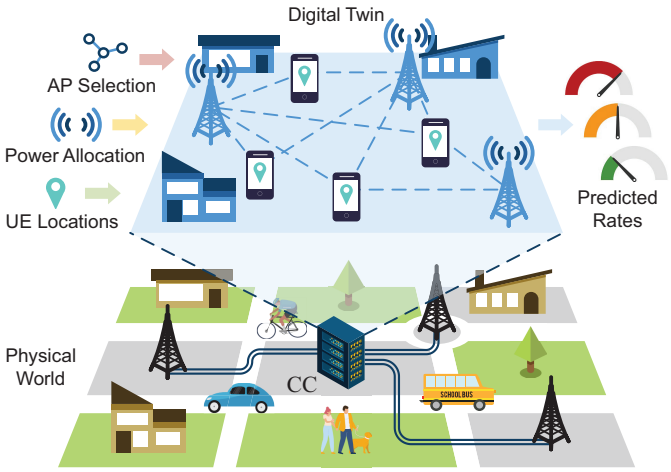


Fig. 2. The constructed digital twin for CF networks.

$$\hat{\mathbf{r}} = \mathcal{G}(\mathbf{X}, \mathbf{B}, \mathbf{P}; \theta), \tag{12}$$

where $\mathcal{G}(\cdot)$ is a neural network model based on graph Transformer [16], and θ denotes the set of its parameters. The parameters θ are learned by minimizing the mean absolute error (MAE) between the predicted and ground-truth rates as

$$\mathcal{L}_{\mathcal{G}} = \frac{1}{K}\sum_{k=1}^{K}\left|R_k - \hat{R}_k\right|. \tag{13}$$

After the optimal parameters θ^* are obtained through back propagation, the constructed model can simultaneously support optimization and verification with the frozen parameters θ^*. Specifically, for optimization, the constructed model is differentiable with respect to the power allocation matrix $\mathbf{P}$. Then, the agent can construct the loss function according to the specified optimization objective. Based on the constructed loss function $\mathcal{L}_R$, the agent then optimizes the power allocation through gradient descent, which is expressed as

$$\mathbf{P}_{n+1} = \mathbf{P}_n - \eta_{\mathbf{P}} \cdot \frac{\partial \mathcal{L}_R}{\partial \mathbf{P}_n}, \tag{14}$$

where $\mathbf{P}_n$ denotes the power allocation matrix at the n-th iteration, and $\eta_{\mathbf{P}}$ is the step size of gradient descent for updating the power allocation matrix. After the predefined number of iterations is reached or convergence is achieved, the optimized strategy with $\mathbf{P}^*$ is passed to the execution environment for verification by

$$\hat{\mathbf{r}}^* = \mathcal{G}(\mathbf{X}, \mathbf{B}, \mathbf{P}^*; \theta^*), \tag{15}$$

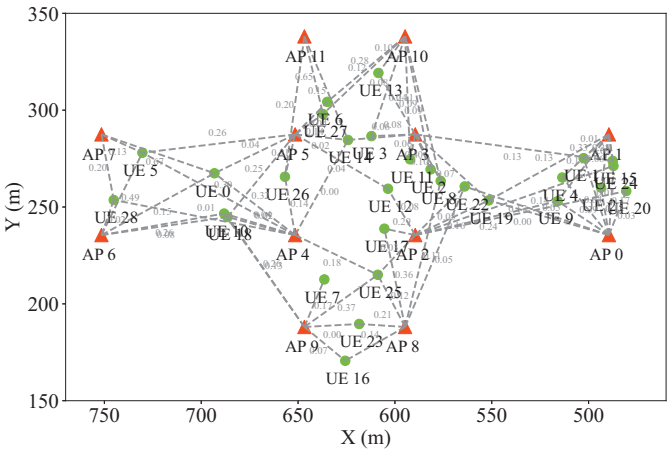


Fig. 3. A generated realization of the CF network.

where $\hat{\boldsymbol{r}}^* = [\hat{R}_1^*, \cdots, \hat{R}_K^*]^T \in \mathbb{R}^{K\times 1}$ denotes the evaluated rates of all K UEs under the candidate strategy $\mathbf{P}^*$. Then, both $\hat{\boldsymbol{r}}^*$ and $\mathbf{P}^*$ are returned to the LLM for further assessment. The LLM compares $\hat{\mathbf{r}}^*$ with the parsed target rate $\boldsymbol{r}^*$ and verifies whether the strategy $\mathbf{P}^*$ satisfies the parsed constraints $\mathcal{C}$. Specifically, if $\hat{R}_k^* \geq R_k^*$, $\forall k = 1, \cdots, K$, and the constraints $\mathcal{C}$ are all satisfied, the strategy $\mathbf{P}^*$ is accepted and output as the executable strategy. Otherwise, the LLM analyzes the specific unsatisfied requirements or violated constraints, updates the task description or optimization configuration based on the feedback, and initiates another optimization loop.

If the task parameter w indicates that joint optimization is required, the optimization tool needs to update both the association matrix $\mathbf{B}$ and the power allocation matrix $\mathbf{P}$. However, joint optimization is challenging for two main reasons. On the one hand, the two variables are tightly coupled. Specifically, $\mathbf{B}$ determines the service topology between APs and UEs, thereby constraining the feasible space of the power allocation matrix $\mathbf{P}$. On the other hand, unlike the continuous power variable $\mathbf{P}$, $\mathbf{B}$ is a binary matrix whose elements can only take values of 0 or 1. This discreteness makes the update of $\mathbf{B}$ discontinuous and difficult to optimize. To address this issue, we first relax $\mathbf{B}$ from the discrete $0, 1$ space to the continuous interval $[0, 1]$, such that it can be treated as a differentiable variable and fed into the constructed differentiable digital twin together with $\mathbf{P}$. Then, a leaky straight-through (ST) estimator [17] is adopted to approximate the backward propagation through the binarization process of $\mathbf{B}$. Specifically, in the forward pass, the relaxed association variable is binarized to obtain an executable topology. In the backward pass, the leaky ST estimator provides an approximate gradient for the binarization operation, allowing the gradient to be propagated from the loss function back to the association variable.

IV. Numerical Simulations and Results

In this section, we introduce the adopted dataset and specify the parameters, followed by simulation results for common and customized optimization tasks of WirelessNetAgent.

A. Dataset and Parameters

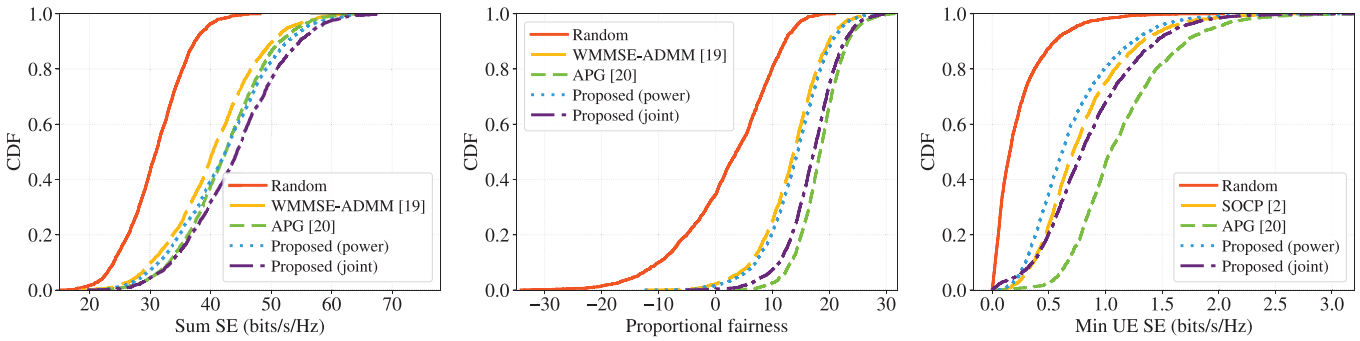
The DeepMIMO [18] 'Outdoor1 (O1)' scenario is adopted in our simulations, and we select a sub-area of the 'O1' scenario comprising 12 APs. To provide an intuitive visualization of the considered CF scenario, Fig. 3 depicts one of the generated realizations of the considered CF network. In this realization, 12 APs jointly serve 29 UEs. The dashed lines indicate the AP-UE associations, while the numbers annotated along the links represent the normalized transmit power. The communication parameters are configured in accordance with a typical 5G new radio (NR) sub-6GHz setting. The carrier frequency is set to $f_c = 3.5$GHz, and a system bandwidth of 10MHz is considered. The subcarrier spacing is set to 15kHz, and 52 resource blocks (RBs) are allocated, each consisting of 12 consecutive subcarriers in the frequency domain. Accordingly, a total of $M = 624$ subcarriers are assigned, occupying an effective bandwidth of 9.36MHz. The remaining 0.64MHz is reserved as guard bands at the spectrum edges to mitigate interference. Each AP is equipped with a UPA comprising $8 \times 4 = 32$ transmit antennas, while each UE is assumed to have a single receive antenna.

B. Common Optimization tasks

The performance of the WirelessNetAgent under three common optimization objectives is reported in Fig. 4, including sum SE maximization, proportional fairness maximization, and minimum user rate maximization. In particular, the random scheme adopts random AP-UE association and power allocation, serving as a reference baseline here. Among the remaining baselines, the weighted minimum mean square error (WMMSE) with alternating direction method of multipliers (ADMM) [19] optimizes only the power allocation and is compared with the proposed power optimization under sum-rate maximization and proportional fairness. Since WMMSE-ADMM is not formulated for the max-min objective, a bisection-based second-order cone programming (SOCP) method [2] is adopted for comparison under this objective. In contrast, accelerated projected gradient (APG) [20] jointly optimizes the AP-UE association and power allocation and is compared with the proposed joint optimization. These conventional methods require instantaneous CSI or channel statistics, whereas the proposed pipeline requires neither as an input during online resource allocation. Despite this inherent information disadvantage, the proposed method remains highly competitive with these CSI-dependent optimization methods as shown in Fig. 4.

C. Customized Optimization tasks

The main advantage of the developed WirelessNetAgent lies in its ability to optimize diverse supported resource-allocation objectives expressed in natural language. For example, given a natural language instruction such as *Improve the rates of users 1, 3, 5 by 50%, and those of users 2, 4, 6 by 20%*, WirelessNetAgent can automatically generate a feasible resource allocation strategy and report the corresponding expected results, as shown in Fig. 5. Due to page limitations, we



(a) Sum-rate maximization (b) Proportional fairness maximization (c) Min-rate maximization

Fig. 4. CDF comparisons under three common optimization objectives. All curves are obtained using rates computed from the DeepMIMO channels.

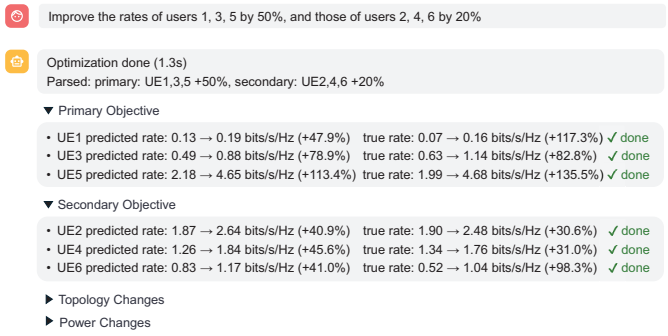


Fig. 5. Interface output of WirelessNetAgent for the example input. Predicted rates are provided by the digital twin, whereas true rates are computed from the DeepMIMO channels using the resulting resource allocation.

are unable to report more cases in this paper. We strongly encourage interested readers to visit our project website[1], where WirelessNetAgent can be tested for customized optimization tasks through natural language interaction.

V. Conclusion

In this paper, we developed WirelessNetAgent, an LLM-empowered AI agent for resource allocation in CF networks. Different from existing model-centric paradigms, WirelessNetAgent adopts a harness engineering approach, where a dedicated support system is built around a general-purpose LLM to guide intent parsing, optimization tool invocation, and execution environment verification. The proposed framework shows that, with proper harness support, general-purpose LLMs can be transformed into capable agents for wireless network optimization, highlighting the great potential in next-generation wireless networks.

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